



Frailty is a Predictor of Increased Readmissions and Increased Postoperative Complications After Elective Treatment of Unruptured Aneurysms

Shivani D. Rangwala¹, Jane S. Han¹, Krista Lamorie-Foote¹, Li Ding², Steven L. Giannotta¹, Frank J. Attenello¹, William Mack¹

INTRODUCTION: Frailty is a state of decreased physiologic reserve associated with adverse treatment outcomes across surgical specialties. We sought to determine whether frailty affected patient outcomes after elective treatment (open microsurgical clipping or endovascular therapy [EVT]) of unruptured cerebral aneurysms (UCAs).

METHODS: The National Readmissions Database was queried from 2010 to 2014 to identify patients who had a known UCA and underwent elective clipping or EVT. Frailty was assessed using the Johns Hopkins Adjusted Clinical Groups frailty indicator tool. Multivariable exact logistic regression analyses were conducted to assess the associations between frailty and the primary outcome variables of 30- and 90-day readmissions, complications, length of stay (LOS), and patient disposition.

RESULTS: Of 18,483 patients who underwent elective treatment for UCAs, 358 (1.9%) met the criteria for frailty. After adjusting for patient- and hospital-based factors, frailty (30-day: odds ratio [OR], 1.55; 95% confidence interval [CI], 1.11–2.17; $P = 0.01$; 90-day: OR, 1.47; 95% CI, 1.05–2.06; $P = 0.02$) and clipping versus EVT (30-day: OR, 2.12; 95% CI, 1.85–2.43; $P < 0.0001$; 90-day: OR, 1.80; 95% CI, 1.59–2.03; $P < 0.0001$) were associated with increased readmission rates. Furthermore, frailty was associated with an

increased rate of complications (surgical: OR, 2.91; 95% CI, 2.27–3.72; $P < 0.0001$; neurological: OR, 3.04; 95% CI, 2.43–3.81; $P < 0.0001$; major: OR, 2.75; 95% CI, 1.96–3.84; $P < 0.0001$), increased LOSs (incidence rate ratio, 3.08; 95% CI, 2.59–3.66; $P < 0.0001$), and an increased rate of nonroutine disposition (OR, 3.94; 95% CI, 2.91–5.34; $P < 0.0001$).

CONCLUSIONS: Frailty was associated with an increased likelihood of 30- and 90-day readmissions after elective treatment of UCAs. Frailty was notably associated with several postoperative complications, longer LOSs, and nonroutine disposition in the treatment of UCAs.

INTRODUCTION

Unruptured cerebral aneurysms (UCAs) have a prevalence of 3.2% and occur in patients at a mean age of 50 years.^{1,2} Rupture risk is influenced by aneurysm size, location, and morphology and patient lifestyle factors.^{3–5} Cerebral aneurysm rupture with resultant subarachnoid hemorrhage (SAH) carries high morbidity and mortality, with mortality rates of 50%.⁶ UCAs could warrant elective treatment to avoid the potential for rupture and its associated complications. The treatment options

Key words

- Clip ligation
- Craniotomy
- Endovascular coil embolization
- Frailty
- Unruptured aneurysm

Abbreviations and Acronyms

APR-DRG: All Patient Refined Diagnosis Related Groups
CCI: Charlson comorbidity index
CI: Confidence interval
EVT: Endovascular therapy
ICD-9: International Classification of Diseases, ninth revision
ICD-10: International Classification of Diseases, tenth revision
IRR: Incidence rate ratio
ISAT: International subarachnoid aneurysm trial
JHACG: Johns Hopkins Adjusted Clinical Groups
LOS: Length of stay

mFI: Modified frailty index

NRD: Nationwide Readmission Database

OR: Odds ratio

SAH: Subarachnoid hemorrhage

UCA: Unruptured cerebral aneurysm

From the ¹Department of Neurosurgery, The Keck School of Medicine of the University of Southern California; and ²Department of Population and Public Health Sciences, Keck School of Medicine, University of Southern California, Los Angeles, California, USA

To whom correspondence should be addressed: Jane S. Han, M.S.
 [E-mail: hansungm@usc.edu]

Citation: World Neurosurg. (2024) 181:e882–e896.

<https://doi.org/10.1016/j.wneu.2023.11.005>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

1878-8750/\$ - see front matter © 2023 Elsevier Inc. All rights reserved.

for UCAs include open microsurgical clipping and endovascular therapy (EVT) with coiling and/or stenting.²

Because the decision to treat is elective, preoperative risk factors should be identified and optimized before intervention. Frailty is a state of decreased physiologic reserve that compromises the ability to cope with stressors.^{7,8} Frailty encompasses physical, psychological, and social factors that can affect an individual's vulnerability to stress.⁹ Previous studies have demonstrated that frailty is associated with an increased incidence of postoperative complications and poor outcomes across multiple surgical subspecialties.¹⁰⁻¹⁶ In neurosurgery, frailty is a predictor of adverse surgical outcomes after specific procedures and for certain cranial pathologies.¹⁷⁻²⁶ To the best of our knowledge, the association between frailty and patient readmissions after elective clipping or EVT for UCAs has not previously been evaluated in the literature.

The present study used the National Readmissions Database (NRD) from 2010 to 2014 to evaluate the effect of a comorbid frailty diagnosis on the primary outcome of patient readmission for patients undergoing treatment for UCAs. We hypothesize that frailty would be associated with increased readmission rates, an increased incidence of in-hospital complications and nonroutine discharge disposition, the overall length of stay (LOS), and the use of microsurgical clipping or EVT for UCAs.

METHODS

Data Acquisition

The Healthcare Cost and Utilization Project's NRD is a public database encompassing inpatient discharge information from 28 states, irrespective of payer option, which accounts for ~60% of hospitalizations in the United States. Patients are assigned unique linkage numbers and tracked within each state throughout 1 calendar year (January to December). All patients undergoing treatment of a UCA provided written informed consent in accordance with university protocol. Institutional review board approval was not needed for any single patient data included in this report.

Study Population

The NRD was queried from 2010 to 2014. Patients undergoing treatment of a UCA were selected according to the International Classification of Diseases, ninth revision (ICD-9), diagnosis and procedure codes. Only ICD-9 was used to reduce data variability and the loss of data during the transition from ICD-9 to ICD, 10th revision (ICD-10). Only the management of UCAs by microsurgical clipping or EVT (i.e., coiling, stenting) was studied. Relevant ICD-9 diagnosis and procedure codes are summarized in [Supplementary Table 1](#). For the patients who underwent elective treatment of a UCA, the exclusion criteria were as follows: age <18 years, death during the index admission, missing survival data, missing LOS data, a diagnosis code of an SAH, and readmission with an unruptured aneurysm. We excluded those with SAH because this population will undergo nonelective treatment of an aneurysm, and the treatment options and outcomes could differ from those for elective treatment. We also excluded those with readmission with an unruptured aneurysm to eliminate patients who might undergo multiple staged treatments (e.g., related to an arteriovenous malformation or

multiple intracranial aneurysms). A diagram showing identification of the study population is presented in [Figure 1](#).

Study Variables

Frailty was defined using the Johns Hopkins Adjusted Clinical Groups (JHACG) method, as described extensively in previous studies.^{20,22,27} The JHACG tool uses ICD-derived diagnoses from 10 clinical elements of frailty: malnutrition, dementia, severe vision impairment, decubitus ulcer, urinary incontinence, fecal incontinence, weight loss, difficulty in walking, falls, and social support needs. Any single diagnosis will characterize a patient as frail. We did not identify the degree of frailty with a greater degree of diagnosis. Frailty was only defined as binary in our analysis. A full list of the frailty diagnoses and ICD-9 codes are listed in [Supplementary Table 2](#).

Outcome Measures

The primary outcomes assessed included readmissions, surgical complications, neurological complications, major hospital complications, and discharge disposition. Readmissions were analyzed at 30 and 90 days. The 30-day cohort was formed from patients discharged from January through November. The 90-day cohort was formed from patients discharged from January through September. Surgical complications included intracranial hemorrhage, intracranial vascular complications, iatrogenic vascular infarction or hemorrhage, mental status changes, delirium, alterations of consciousness or an altered state, hematoma complicating the procedure, postoperative nausea with or without vomiting, and bacterial meningitis. Neurological complications included transient cerebral ischemia, SAH, seizures, and neurological complications after the procedure. Major hospital complications included deep vein thrombosis, pulmonary embolism, pneumonia, renal failure, sepsis or septic shock, stroke, and cardiac arrest. The specific diagnoses and ICD-9 codes for all complications are presented in [Supplementary Table 3](#). The hospital outcome measures included the LOS and discharge disposition (i.e., routine vs. nonroutine). Routine discharges are characterized by discharge to home. Nonroutine discharges include inpatient rehabilitation, a requirement for home health services, discharge to a skilled nursing facility, and discharge to other intermediate- or long-term care facilities.

Statistical Analysis

Patient demographics and clinical characteristics are reported as frequencies and percentages stratified by frailty status. Differences between the 2 groups were evaluated using χ^2 tests. Multivariable regression analyses were conducted, adjusting for covariates and using generalized estimation equations to account for hospital clustering. Logistic regression was used for binary outcomes. Results are reported as odds ratios (ORs). The Hosmer-Lemeshow goodness-of-fit test was used to check model assumptions. Negative binomial regression was used for the LOS outcome, adjusting for variance inflation. The results are reported as incident rate ratios (IRRs). Covariates in the final model included the previously cited study variables. For 30- and 90-day readmissions, complications and discharge disposition were considered as mediators and were not included in the model. The incidence of missing data was <2%, and only complete cases were used for

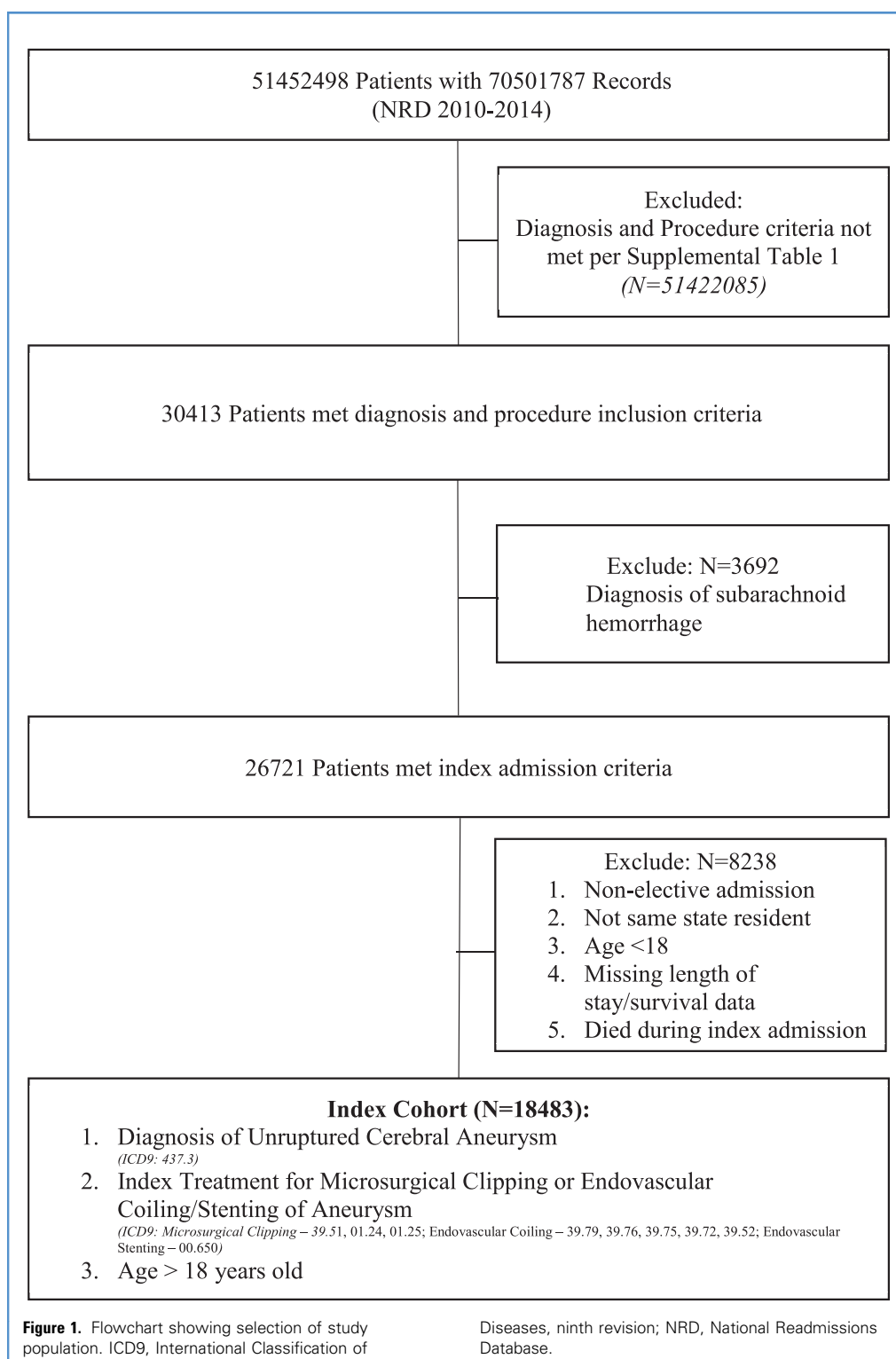


Table 1. Patient and Hospital Demographics Stratified by Frailty

Variable	Frailty (n, %)		P Value
	Yes	No	
Total	358 (1.9)	18,125 (98.1)	
Treatment group			< 0.0001*
Clipping	184 (51.4)	5715 (31.5)	
Coiling/stenting	174 (48.6)	12,410 (68.5)	
Age (years)			< 0.0001*
≤44	33 (9.2)	2397 (13.2)	
45–59	128 (35.8)	7285 (40.2)	
60–74	147 (41.1)	7079 (39.1)	
>75	50 (14.0)	1364 (7.5)	
Sex			0.008*
Male	105 (29.3)	4228 (23.3)	
Female	253 (76.7)	13,897 (76.7)	
Insurance			< 0.0001*
Missing	DS	36 (0.2)	
Medicare	172 (48.0)	6797 (37.5)	
Medicaid	49 (13.7)	2084 (11.5)	
Private insurance	111 (31.0)	8002 (44.2)	
Other	25 (7.0)	1206 (6.7)	
APR-DRG risk of mortality subclass			< 0.0001*
No class	DS	DS	
Minor	193 (54.0)	14,994 (82.7)	
Moderate	85 (23.7)	2180 (12.0)	
Major	42 (11.7)	514 (2.8)	
Extreme	38 (10.6)	435 (2.4)	
APR-DRG severity of illness subclass			< 0.0001*
No class	DS	DS	
Minor	40 (11.2)	7468 (41.2)	
Moderate	156 (43.6)	8477 (46.8)	
Major	113 (31.6)	1745 (9.6)	
Extreme	49 (13.7)	433 (2.4)	
LOS (days)			< 0.0001*
≤1	54 (15.1)	7799 (43.0)	
2	44 (12.3)	3612 (20.0)	
3–4	53 (14.8)	3757 (20.7)	
≥5	207 (57.8)	2957 (16.3)	
CCI			< 0.0001*
2	155 (43.3)	11,350 (62.2)	
≥3	203 (56.7)	6775 (37.4)	

DS, data suppressed in accordance with HCUP data user agreement regarding cell values <10; APR-DRG, all patient refined diagnostic related groups; LOS, length of stay; CCI, charlson comorbidity index; HCUP, healthcare cost and utilization project.

*Statistically significant.

†Calculated using the patient's ZIP Code, based on current year.

Continues

Table 1. Continued

Variable	Frailty (<i>n</i> , %)		<i>P</i> Value
	Yes	No	
Median household income†			0.02*
Missing	DS	336 (1.9)	
0–25th percentile	97 (27.1)	4119 (22.7)	
26th–50th percentile	89 (24.9)	4199 (23.2)	
51st–75th percentile	94 (26.3)	4714 (26.0)	
76th–100th percentile	69 (19.3)	4757 (26.3)	
HCUP used emergency			0.001*
No	351 (98.0)	18,015 (99.3)	
Yes	DS	110 (0.6)	
Discharge disposition			< 0.0001*
Missing	DS	DS	
Routine	209 (58.4)	15,979 (88.1)	
Nonroutine	149 (41.6)	2143 (11.8)	
Major complication			< 0.0001*
No	304 (85.0)	17,411 (96.0)	
Yes	54 (15.0)	714 (4.0)	
Surgical complication			< 0.0001*
No	262 (73.2)	16,415 (90.6)	
Yes	96 (26.8)	1710 (9.4)	
Neurological complication			< 0.0001*
No	261 (72.9)	16,438 (90.7)	
Yes	97 (27.1)	1687 (9.3)	
Teaching status			0.41
Nonteaching hospital	29 (8.1)	1700 (9.4)	
Teaching hospital	329 (91.9)	16,425 (90.6)	
Metropolitan location			0.48
Other	107 (30.0)	5733 (31.6)	
Major metropolitan area	251 (70.0)	12,392 (68.4)	
Hospital bed size			0.71
Small	20 (5.6)	1007 (5.6)	
Medium	42 (11.7)	2394 (13.2)	
Large	296 (82.7)	14,724 (81.2)	
Hospital control			0.4
Government, nonfederal	82 (22.9)	3827 (21.1)	
Private	276 (77.1)	14,298 (78.9)	
Hospital volume			0.32
≤90th percentile (117/year)	85 (23.7)	3908 (21.6)	
>90th percentile	273 (76.3)	14,217 (78.4)	

DS, data suppressed in accordance with HCUP data user agreement regarding cell values <10; APR-DRG, all patient refined diagnostic related groups; LOS, length of stay; CCI, charlson comorbidity index; HCUP, healthcare cost and utilization project.

*Statistically significant.

†Calculated using the patient's ZIP Code, based on current year.

multivariable regression. $P < 0.05$ was considered statistically significant. Analysis was performed using SAS, version 9.4 (SAS Institute, Cary, North Carolina, USA).

RESULTS

Patient Demographics

A total of 18,483 patients underwent elective treatment for a known UCA. Of these patients, 358 (1.9%) met the criteria for frailty using the JHACG tool. A greater proportion of frail patients were aged >60 years (55.1% vs. 46.6%; $P < 0.0001$) and were more likely to be male (29.3% vs. 23.3%; $P < 0.008$) compared with the nonfrail patients. A greater proportion of frail patients underwent microsurgical clipping than did the nonfrail patients (51.4% vs. 31.5%; $P < 0.0001$). Also, more frail patients were insured with Medicare or Medicaid (61.7% vs. 49.0%; $P < 0.0001$). Frail patients were more often classified in a moderate to extreme All Patient Refined Diagnosis Related Groups (APR-DRG) mortality subclass (46.0% vs. 17.2%; $P < 0.0001$) and a major to extreme APR-DRG illness severity subclass (45.3% vs. 12.0%; $P < 0.0001$). A greater number of frail patients had ≥ 3 comorbid conditions using the Charlson comorbidity index (CCI; 56.7% vs. 37.3%; $P < 0.0001$). The frail cohort was more likely to require a nonroutine disposition (41.6% vs. 11.8%; $P < 0.0001$) and to have a prolonged hospital LOS (≥ 5 days; 57.8% vs. 16.3%; $P < 0.0001$). Frail patients also experienced more surgical (26.8% vs. 9.4%; $P < 0.0001$), neurological (27.1% vs. 9.3%; $P < 0.0001$), and major hospital (15.1% vs. 3.9%; $P < 0.0001$) complications. The patient demographics are summarized in [Table 1](#).

Effect of Frailty on Patient Outcomes

After adjusting for patient- and hospital-based factors, multivariate analyses evaluating the effect of frailty on patient outcomes (i.e., complication rates, discharge disposition, LOS, and readmissions) were conducted. Frailty was significantly associated with an increased likelihood of readmission within 30 (OR, 1.55; 95% CI, 1.11–2.17; $P = 0.01$) or 90 (OR, 1.47; 95% CI, 1.05–2.06; $P = 0.02$) days. Microsurgical clipping (30-day: OR, 2.12; 95% CI, 1.85–2.43; $P < 0.0001$; 90-day: OR, 1.80; 95% CI, 1.59–2.03; $P < 0.0001$) and a higher CCI (30-day: OR, 1.66; 95% CI, 1.45–1.89; $P < 0.0001$; 90-day: OR, 1.70; 95% CI, 1.51–1.91; $P < 0.0001$) were associated with an increased odds of readmission at 30 and 90 days. The factors associated with a greater likelihood of readmission within 30 and 90 days are summarized in [Table 2](#). The most common causes for 30- and 90-day readmission after microsurgical clipping were postoperative infection and subdural hemorrhage. For EVT, stroke, transient ischemic attack, and headaches were the most common reasons. Other causes for 30- and 90-day readmission are summarized in [Table 3](#).

The frail cohort was associated with an increased likelihood of experiencing surgical (OR, 2.91; 95% CI, 2.27–3.72; $P < 0.0001$), neurological (OR, 3.04; 95% CI, 2.43–3.81; $P < 0.0001$), and major hospital (OR, 2.75; 95% CI, 1.96–3.84; $P < 0.0001$) complications. Patients who underwent microsurgical clipping also had a greater likelihood of experiencing surgical (OR, 1.94; 95% CI, 1.72–2.19; $P < 0.0001$), neurological (OR, 1.37; 95% CI, 1.22–1.54; $P < 0.0001$), and major hospital (OR, 2.96; 95% CI, 2.50–3.49; $P < 0.0001$) complications. A CCI of >3 was also associated

with an increased odds of ≥ 1 complications. Significant associations with postoperative complications on multivariate analysis are summarized in [Table 4](#).

Frailty was associated with an increased likelihood of requiring a nonroutine disposition (OR, 3.94; 95% CI, 2.91–5.34; $P < 0.0001$). Other factors associated with a higher likelihood of nonroutine disposition were microsurgical clipping (OR, 6.01; 95% CI, 5.29–6.83; $P < 0.0001$) and an increased number of comorbidities (OR, 1.94; 95% CI, 1.77–2.13; $P < 0.0001$). Frailty was also associated with increased likelihood of a prolonged hospital LOS (IRR, 3.08; 95% CI, 2.59–3.66; $P < 0.0001$). Additional factors associated with an increased likelihood of a prolonged hospital LOS were microsurgical clipping (IRR, 2.88; 95% CI, 2.73–3.05; $P < 0.0001$) and multiple comorbidities (IRR, 1.45; 95% CI, 1.38–1.51; $P < 0.0001$). Significant associations with a nonroutine discharge disposition and a prolonged hospital LOS are presented in [Table 5](#).

DISCUSSION

The NRD from 2010 to 2014 was used to investigate the role of frailty for 18,483 patients who underwent elective treatment for UCAs. To the best of our knowledge, this is the first study using a national dataset to evaluate the association between frailty and readmissions after treatment of UCAs. Using the JHACG frailty index, the incidence of frailty was 1.9% among this identified study population. We found frailty to be significantly associated with an increased likelihood of both 30- and 90-day readmissions. In addition, frailty was associated with an increased likelihood of surgical, neurological, and major hospital complications.

In the current literature, the reported frailty incidence is 1.48%–7.3% among cranial neurosurgery patients, depending on the patient cohort and frailty metric.^{17,19,20,22,23,28,29} Other studies have shown frailty to be a predictor of increased unplanned readmissions within various surgical specialties^{30–33}; however, the effect of frailty on unplanned readmissions after elective treatment of UCAs has not been studied. We found frailty to be associated with an increased risk of readmissions at 30 and 90 days. Furthermore, our findings showed a higher risk of readmissions for frail patients undergoing surgical clipping compared with those who underwent EVT, which has also not been previously studied. A NRD study found microsurgical clipping to significantly increase the likelihood of 30-day readmission compared with EVT for UCAs.³⁴ That study, however, did not directly assess frailty.³⁴ A prior national database study, although again not directly assessing frailty, discovered that a greater comorbidity burden (CCI ≥ 3) was associated with increased 30- and 90-day readmissions for patients undergoing elective EVT of UCAs.³² More recent studies have found frailty to be a better metric for patient assessment than age³⁵ and that frailty significantly increased the risk of poor postoperative outcomes after treatment of anterior circulation unruptured aneurysms.³⁶ In the treatment of UCAs in the elderly, the physiologic reserve is already decreased compared with that of younger patients, and frailty can serve as a direct assessment of this for treatment planning.³⁷ In addition to aneurysm morphology and location, frailty should be included in the preoperative assessment for elective

Table 2. Predictors of 30- and 90-Day All-Cause Readmissions

Variable	30-Day Readmissions		90-Day Readmissions	
	OR (95% CI)	P Value	OR (95% CI)	P Value
Frailty				
Yes	1.55 (1.11–2.17)	0.01*	1.47 (1.05–2.06)	0.02*
No	Ref	—	Ref	—
Treatment group				
Clipping	2.12 (1.85–2.43)	< 0.0001*	1.80 (1.59–2.03)	< 0.0001*
Coiling/stenting	Ref	—	Ref	—
Age (years)				
≤44	Ref	—	Ref	—
45–59	1.09 (0.86–1.38)	0.46	1.10 (0.88–1.37)	0.397
60–74	0.99 (0.77–1.27)	0.95	1.05 (0.83–1.33)	0.68
>75	1.30 (0.94–1.81)	0.12	1.55 (1.14–2.09)	0.005*
Sex				
Male	1.26 (1.08–1.46)	0.0034*	1.17 (1.03–1.35)	0.021*
Female	Ref	—	Ref	—
Insurance				
Medicare	1.20 (0.94–1.54)	0.14	1.18 (0.93–1.48)	0.17
Medicaid	Ref	—	Ref	—
Private insurance	0.75 (0.59–0.95)	0.018*	0.73 (0.59–0.89)	0.0024*
Other	0.93 (0.68–1.27)	0.64	0.91 (0.68–1.22)	0.54
CCI				
2	Ref	—	Ref	—
≥3	1.66 (1.45–1.89)	< 0.0001*	1.70 (1.51–1.91)	< 0.0001*
Discharge quarter				
January–March	Ref	—	Ref	—
April–June	0.95 (0.79–1.14)	0.59	0.91 (0.79–1.06)	0.21
July–September	0.94 (0.78–1.12)	0.49	0.96 (0.84–1.11)	0.58
October–November	0.86 (0.70–1.06)	0.17	—	—
Median household income†				
0–25th percentile	Ref	—	Ref	—
26th–50th percentile	0.90 (0.74–1.10)	0.30	0.97 (0.82–1.16)	0.77
51st–75th percentile	0.94 (0.78–1.12)	0.47	0.95 (0.81–1.12)	0.55
76th–100th percentile	0.87 (0.71–1.05)	0.15	0.89 (0.75–1.07)	0.21
Hospital volume				
≤90th percentile (117/year)	0.86 (0.71–1.05)	0.15	0.86 (0.72–1.03)	0.09
>90th percentile	Ref	—	Ref	—
Hospital control				
Government, nonfederal	Ref	—	Ref	—
Private	0.94 (0.80–1.11)	0.48	0.93 (0.80–1.08)	0.34

Continues

Table 2. Continued

Variable	30-Day Readmissions		90-Day Readmissions	
	OR (95% CI)	P Value	OR (95% CI)	P Value
Hospital bed size				
Small	0.80 (0.58–1.10)	0.16	0.87 (0.66–1.15)	0.33
Medium	1.03 (0.82–1.28)	0.81	1.10 (0.93–1.31)	0.26
Large	Ref	—	Ref	—
Teaching status				
Nonteaching hospital	1.10 (0.84–1.42)	0.50	0.96 (0.76–1.22)	0.75
Teaching hospital	Ref	—	Ref	—
Metropolitan location				
Other	0.91 (0.79–1.05)	0.199	1.00 (0.89–1.13)	0.98
Major metropolitan	Ref	—	Ref	—

OR, odds ratio; CI, confidence interval; Ref, referent; CCI, charlson comorbidity index.
 *Statistically significant.
 †Calculated using the patient's ZIP Code, based on current year.

treatment options for UCAs to better assess the risks involved with each treatment option. Interventional goals can be more judiciously aligned to minimize the surgical and anesthetic times and reduce the overall invasiveness of the procedure (clipping vs. EVT). Clipping was associated with an increased likelihood of worse patient outcomes in our study, suggesting that frailty scores should be more carefully considered when deciding candidacy for this type of procedure. The

preoperative workup could also prioritize patient prehabilitation for intervention to optimize outcomes in the postoperative period for this high-risk patient population. Implementation of prehabilitation and enhanced recovery pathway programs has been shown to improve surgical outcomes for frail patients.^{38,39} Furthermore, patients were most commonly readmitted for infection, subdural hemorrhage, transient ischemic attack or stroke, and headaches in our

Table 3. Causes of 30- and 90-Day Readmissions

Variable	30-Day Readmissions		90-Day Readmissions	
	Clipping	Coiling/Stenting	Clipping	Coiling/Stenting
Postoperative infection	28 (6.02)	—	41 (7.41)	—
Cerebral artery occlusion with infarction	17 (3.66)	41 (7.99)	17 (3.07)	55 (7.58)
Subdural hemorrhage	19 (4.09)	—	40 (7.23)	—
Headache	19 (4.09)	23 (4.48)	17 (3.07)	23 (3.17)
Transient cerebral ischemia	—	—	—	23 (3.17)
Intracerebral hemorrhage	—	15 (2.92)	—	—
Cerebral embolism with infarction	—	15 (2.92)	—	—
Subarachnoid hemorrhage	—	14 (2.73)	—	—
Urinary tract infection	—	—	—	22 (3.03)
Septicemia	15 (3.23)	—	—	16 (2.2)
Other nervous system complications	—	—	17 (3.07)	—

Data presented as n (%).

Table 4. Postoperative Complications

Variable	Surgical Complication		Neurological Complication		Major Complication	
	OR (95% CI)	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Frailty						
Yes	2.91 (2.27–3.72)	< 0.0001*	3.04 (2.43–3.81)	< 0.0001*	2.75 (1.96–3.84)	< 0.0001*
No	Ref	—	Ref	—	Ref	—
Treatment group						
Clipping	1.94 (1.72–2.19)	< 0.0001*	1.37 (1.22–1.54)	< 0.0001*	2.96 (2.50–3.49)	< 0.0001*
Coiling/stenting	Ref	—	Ref	—	Ref	—
Age (years)						
≤44	Ref	—	Ref	—	Ref	—
45–59	1.12 (0.94–1.33)	0.20	1.02 (0.86–1.21)	0.84	1.16 (0.87–1.56)	0.30
60–74	1.31 (1.08–1.59)	0.0058*	1.02 (0.85–1.23)	0.84	1.76 (1.28–2.42)	0.0005*
>75	1.57 (1.23–2.0)	0.0003*	1.04 (0.82–1.33)	0.73	2.13 (1.43–3.17)	0.0002*
Sex						
Male	0.89 (0.79–1.0)	0.066	1.12 (0.998–1.26)	0.055	1.52 (1.29–1.79)	< 0.0001*
Female	Ref	—	Ref	—	Ref	—
CCI						
2	Ref	—	Ref	—	Ref	—
≥3	1.42 (1.29–1.58)	< 0.0001*	1.52 (1.37–1.69)	< 0.0001*	3.06 (2.61–3.60)	< 0.0001*
Insurance						
Medicare	1.01 (0.84–1.22)	0.90	0.87 (0.73–1.03)	0.104	0.95 (0.71–1.26)	0.72
Medicaid	Ref	—	Ref	—	Ref	—
Private insurance	0.90 (0.77–1.06)	0.22	0.65 (0.56–0.76)	< 0.0001*	0.82 (0.63–1.07)	0.15
Other	0.96 (0.75–1.2)	0.77	0.76 (0.59–0.97)	0.028*	0.74 (0.49–1.11)	0.15
Median household income†						
0–25th percentile	Ref	—	Ref	—	Ref	—
26th–50th percentile	1.03 (0.88–1.21)	0.71	1.03 (0.89–1.2)	0.69	1.0 (0.81–1.22)	0.96
51st–75th percentile	1.06 (0.92–1.23)	0.44	0.91(0.79–1.05)	0.19	0.80 (0.64–1.0)	0.055
76th–100th percentile	1.18 (1.01–1.37)	0.043*	1.0 (0.86–1.16)	0.98	0.89 (0.7–1.12)	0.32
Hospital volume						
≤90th percentile	0.97 (0.82–1.15)	0.73	1.07 (0.91–1.26)	0.42	1.37 (1.08–1.73)	0.0083*
>90th percentile	Ref	—	Ref	—	Ref	—
Metropolitan location						
Other	0.75 (0.64–0.87)	0.0002*	0.74 (0.64–0.86)	< 0.0001*	0.76 (0.63–0.92)	0.0057*
Major metropolitan area	Ref	—	Ref	—	Ref	—

OR, odds ratio; CI, confidence interval; Ref, referent; CCI, charlson comorbidity index.
 *Statistically significant.
 †Calculated using the patient's ZIP Code, based on current year.

Table 5. Nonroutine Discharge Disposition and Prolonged Hospital Stay

Variable	Nonroutine Disposition		Length of Stay	
	OR (95% CI)	P Value	IRR (95% CI)	P Value
Frailty				
Yes	3.94 (2.91–5.34)	< 0.0001*	3.08 (2.60–3.66)	< 0.0001*
No	Ref	—	Ref	—
Treatment group				
Clipping	6.01 (5.29–6.83)	< 0.0001*	2.88 (2.73–3.05)	< 0.0001*
Coiling/stenting	Ref	—	Ref	—
Age (years)				
≤44	Ref	—	Ref	—
45–59	1.54 (1.26–1.87)	< 0.0001*	0.97 (0.91–1.03)	0.33
60–74	2.67 (2.19–3.25)	< 0.0001*	1.07 (0.999–1.14)	0.55
>75	5.18 (4.03–6.65)	< 0.0001*	1.27 (1.15–1.41)	< 0.0001*
Sex				
Male	0.88 (0.78–0.99)	0.035*	1.05 (0.999–1.10)	0.054
Female	Ref	—	Ref	—
Insurance				
Medicare	1.11 (0.91–1.34)	0.30	0.95 (0.87–1.03)	0.22
Medicaid	Ref	—	Ref	—
Private insurance	0.59 (0.50–0.71)	< 0.0001*	0.86 (0.80–0.92)	< 0.0001*
Other	0.79 (0.61–1.03)	0.085	0.91 (0.81–1.02)	0.11
CCI				
2	Ref	—	Ref	—
≥3	1.94 (1.77–2.13)	< 0.0001*	1.45 (1.38–1.51)	< 0.0001*
Hospital volume				
≤90th percentile	1.16 (0.98–1.37)	0.077	1.14 (1.05–1.24)	0.002*
>90th percentile	Ref	—	Ref	—
Teaching status				
Nonteaching hospital	1.06 (0.83–1.35)	0.66	1.14 (1.0–1.30)	0.043*
Teaching hospital	Ref	—	Ref	—
Metropolitan location				
Other	0.86 (0.74–1.0)	0.056	0.87 (0.81–0.94)	0.0002*
Major metropolitan area	Ref	—	Ref	—

OR, odds ratio; CI, confidence interval; IRR, incidence rate ratio; Ref, referent; CCI, charlson comorbidity index.

*Statistically significant.

study, suggesting that postoperatively patients should be monitored more extensively and symptom management should be conducted more prudently before discharge for these common causes. Early follow-up of this patient cohort should also be considered, because these patients might be more vulnerable to the fragmentations that exist in transitional care.

Our results characterize frailty as a significant risk factor associated with adverse outcomes for patients undergoing elective treatment for UCAs, which is supported by other studies.^{35,36} Previous studies of various neurosurgical pathologies have shown similar results.^{18,20,22} Asemota and Gallia²⁰ investigated the influence of frailty on postoperative outcomes for patients

who underwent transsphenoidal pituitary surgery using the National (Nationwide) Inpatient Sample with JHACG criteria to define frailty, similar to our study. Our group also previously demonstrated that frailty, when also defined by the JHACG, was associated with an increased incidence of surgical complications, increased LOSs, and an increased incidence of nonroutine disposition for patients undergoing craniotomy for brain tumor resection.²²

Several other instruments exist for diagnosing frailty, including the modified frailty index (mFI).²¹ Despite the variances in diagnostic criteria when determining frailty, studies using other indexes have found similar findings in neurosurgical patients.^{18,19,21,40} In a national database study using the mFI, frailty was a reliable predictor of both medical and neurologic complications, prolonged hospital LOSs, and unfavorable discharge for oncologic neurosurgery patients.¹⁹ Frailty, when defined as the presence of anemia, hypoalbuminemia, or a low body mass index, was previously reported as an independent predictor of poor postoperative outcomes for patients with aneurysmal SAH aged ≥ 60 years.¹⁸ A systematic review demonstrated that frailty was associated with increased postoperative complications and mortality for cranial neurosurgery patients.²¹ Moreover, a meta-analysis across surgical subspecialties found that frailty was associated with surgical complications, major complications (defined as a Clavien-Dindo class $>III$), wound complications, and hospital readmissions. Furthermore, frail patients were more likely to require discharge to a skilled care facility.¹⁰ The mFI was applied to a national Japanese database to study the effects on unruptured aneurysm management and found frailty to be a more accurate predictor of increased in-hospital complications than other patient characteristics.³⁵ Categorizing frailty in a graded fashion (i.e., low-, intermediate-, and high-risk frailty) still resulted in high-risk frailty being a greater predictor of poor outcomes compared with other comorbidities for patients undergoing treatment of UCAs.³⁶ No matter how frailty is assessed, it is clear that frailty is associated with an increased incidence of complications and poor outcomes for patients undergoing treatment of UCAs and should be assessed before elective surgical intervention to optimize patient outcomes.

In the present study, a greater proportion of frail patients underwent microsurgical clipping instead of EVT compared with nonfrail patients. Since the publication of the ISAT (international subarachnoid aneurysm trial), many institutions have adopted a “coil-first” approach.^{41,42} This method reserves surgical clipping for complex aneurysms in locations that prohibit endovascular coiling/stenting, because EVT was found to result in better patient outcomes, such as reduced mortality and dependency at 1 year.^{41,43} Moreover, the International Study of Unruptured Intracranial Aneurysms discovered that age is a predictor of adverse surgical outcomes on multivariate analysis.³ Given these reasons, EVT is generally usually favored for older adults due to the risks associated with open surgery.^{2,44} With such recommendations, this was a noteworthy finding, given that frailty is often associated with older age, because it is thought to be representative of patients’ biological age, and patients who are older are more likely to be in the frail cohort, such

as seen in our study. Predictably, our results showed that microsurgical clipping was associated with an increased likelihood of poor postoperative outcomes. Congruent with our findings, a national database study of patients treated for UCAs found that clipping was associated with an increased likelihood of neurological complications and nonroutine discharge.⁴⁵ It is possible that elective intervention of UCAs in frail patients was reserved for those with more complex aneurysms with a higher rupture risk. The threshold for elective surgical intervention should be higher for frail patients compared with nonfrail patients due to the potential surgical risks and higher risk of unplanned readmissions.

Study Limitations

This study has several limitations. The lack of a standard measure for frailty assessment could have introduced a bias. Many methods have been described, with a single review identifying 67 frailty instruments.⁴⁶ The JHACG and mFI are common measures of frailty used in database studies. The JHACG indicator quantifies frailty using specific diagnosis codes.⁴⁷ The mFI evaluates a selection of frailty-associated conditions, such as hypertension and diabetes.^{19,47} Dementia is the only shared condition between the JHACG tool and the mFI. Different patient populations are assessed with each metric, which could influence the generalizability of the results. Furthermore, frailty is reported as a binary variable using the JHACG method. As such, we were unable to determine the degree of frailty and its effect on surgical outcomes. More research is needed to determine the optimal method to assess and stratify frailty for neurosurgical patients. Further investigations on which comorbidities are most contributory to increasing perioperative risk in this specific patient group should also be conducted. Furthermore, this study queried the NRD, which is subject to the limitations of an administrative database, including information bias and coding errors. Thus, we were unable to investigate specific patient, provider, and hospital factors, such as provider volume and hospital policies and/or practices. Also, due to the data variability between ICD-9 and ICD-10 coding and the loss of data during the transition from ICD-9 to ICD-10, our data were queried only using ICD-9 codes and captured data from 2010 to 2014. Additionally, given that the NRD resets annually, a longitudinal analysis could not be conducted, restricting the patient population available for our study.

CONCLUSIONS

Frailty is independently associated with unplanned readmission at both 30 and 90 days after elective intervention. Frailty was associated with an increased likelihood of surgical, neurological, and major complications on multivariate analysis for both microsurgical clipping and EVT. Frail patients were more likely to have an unplanned readmission at both 30 and 90 days after surgical clipping (relative to EVT). Further studies are required to confirm our findings and potentially to investigate the feasibility of such recovery pathway programs as preoperative interventions to optimize outcomes for frail patients requiring elective surgical

or endovascular interventions for unruptured aneurysm treatment.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Shivani D. Rangwala: Investigation, Writing — original draft, Writing — review & editing, Conceptualization, Visualization. **Jane S. Han:** Investigation, Writing — original draft, Writing — review

& editing, Visualization. **Krista Lamorie-Foote:** Investigation, Writing — original draft. **Li Ding:** Investigation, Data curation, Methodology. **Steven L. Giannotta:** Supervision. **Frank J. Attenello:** Writing — review & editing, Conceptualization, Supervision. **William Mack:** Writing — review & editing, Conceptualization, Supervision.

REFERENCES

1. Vlak MH, Algra A, Brandenburg R, Rinkel GJ. Prevalence of unruptured intracranial aneurysms, with emphasis on sex, age, comorbidity, country, and time period: a systematic review and meta-analysis. *Lancet Neurol.* 2011;10:626-636.
2. Thompson BG, Brown RD, Amin-Hanjani S, et al. Guidelines for the management of patients with unruptured intracranial aneurysms: a Guideline for Healthcare professionals from the American heart association/American stroke association. *Stroke.* 2015;46:2368-2400.
3. Wiebers DO, Whisnant JP, Huston J, et al. Unruptured intracranial aneurysms: natural history, clinical outcome, and risks of surgical and endovascular treatment. *Lancet.* 2003;362:103-110.
4. Raghavan ML, Ma B, Harbaugh RE. Quantified aneurysm shape and rupture risk. *J Neurosurg.* 2005;102:355-362.
5. Korja M, Lehto H, Juvela S. Lifelong rupture risk of intracranial aneurysms depends on risk factors: a prospective Finnish cohort study. *Stroke.* 2014;45:1958-1963.
6. van Gijn J, Kerr RS, Rinkel GJE. Subarachnoid haemorrhage. *Lancet.* 2007;369:306-318.
7. Chen X, Mao G, Leng SX. Frailty syndrome: an overview. *Clin Interv Aging.* 2014;9:433-441.
8. Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci.* 2001;56:M146-M156.
9. Gobbens RJ, van Assen MA, Luijckx KG, Wijnen-Sponselee MT, Schols JMGA. Determinants of frailty. *J Am Med Dir Assoc.* 2010;11:356-364.
10. Panayi AC, Orkaby AR, Sakthivel D, et al. Impact of frailty on outcomes in surgical patients: a systematic review and meta-analysis. *Am J Surg.* 2019;218:393-400.
11. Goel NJ, Iyengar A, Kelly JJ, Han JJ, Brown CR, Desai ND. Volume of frail patients predicts outcome in frail patients after cardiac surgery. *J Thorac Cardiovasc Surg.* 2022;163:151-160.e6.
12. Iyengar A, Goel N, Kelly JJ, et al. Effects of frailty on outcomes and 30-day readmissions after surgical mitral valve replacement. *Ann Thorac Surg.* 2020;109:1120-1126.
13. Leven DM, Lee NJ, Kothari P, et al. Frailty index is a significant predictor of complications and mortality after surgery for adult spinal deformity. *Spine (Phila Pa 1976).* 2016;41:E1394-E1401.
14. Chappidi MR, Kates M, Patel HD, et al. Frailty as a marker of adverse outcomes in patients with bladder cancer undergoing radical cystectomy. *Urol Oncol.* 2016;34:256.e1-256.e6.
15. Arya S, Long CA, Brahmabhatt R, et al. Preoperative frailty increases risk of nonhome discharge after elective vascular surgery in home-dwelling patients. *Ann Vasc Surg.* 2016;35:19-29.
16. Nieman CL, Pitman KT, Tufaro AP, Eisele DW, Frick KD, Gourin CG. The effect of frailty on short-term outcomes after head and neck cancer surgery. *Laryngoscope.* 2018;128:102-110.
17. Shahrestani S, Lechrich BM, Tafreshi AR, et al. The role of frailty in geriatric cranial neurosurgery for primary central nervous system neoplasms. *Neurosurg Focus.* 2020;49:E15.
18. Yue Q, Liu Y, Leng B, et al. A prognostic model for early post-treatment outcome of elderly patients with aneurysmal subarachnoid hemorrhage. *World Neurosurg.* 2016;95:253-261.
19. Youngerman BE, Neugut AI, Yang J, Hershan DL, Wright JD, Bruce JN. The modified frailty index and 30-day adverse events in oncologic neurosurgery. *J Neurooncol.* 2018;136:197-206.
20. Asemota AO, Gallia GL. Impact of frailty on short-term outcomes in patients undergoing transphenoidal pituitary surgery. *J Neurosurg.* 2019;132:360-370.
21. Pazniokas J, Gandhi C, Theriault B, et al. The immense heterogeneity of frailty in neurosurgery: a systematic literature review. *Neurosurg Rev.* 2021;44:189-201.
22. Bonney PA, Chartrain AG, Briggs RG, et al. Frailty is associated with in-hospital morbidity and nonroutine disposition in brain tumor patients undergoing craniotomy. *World Neurosurg.* 2021;146:e1045-e1053.
23. Cloney M, D'Amico R, Lebovic J, et al. Frailty in geriatric Glioblastoma patients: a predictor of operative morbidity and outcome. *World Neurosurg.* 2016;89:362-367.
24. Harland TA, Wang M, Gunaydin D, et al. Frailty as a predictor of neurosurgical outcomes in brain tumor patients. *World Neurosurg.* 2020;133:e813-e818.
25. Imaoka Y, Kawano T, Hashiguchi A, et al. Modified frailty index predicts postoperative outcomes of spontaneous intracerebral hemorrhage. *Clin Neurol Neurosurg.* 2018;175:137-143.
26. Shimizu K, Sadatomo T, Hara T, Onishi S, Yuki K, Kurisu K. Importance of frailty evaluation in the prediction of the prognosis of patients with chronic subdural hematoma. *Geriatr Gerontol Int.* 2018;18:1173-1176.
27. The Johns Hopkins ACG® system version 11.0 technical. Available at: <https://studylib.net/doc/18796198/the-johns-hopkins-acg-system-version-11.0-technical>. Accessed January 16, 2023.
28. Shahrestani S, Ballatori AM, Chen XT, et al. Analysis of modifiable and nonmodifiable risk factors in patients undergoing pituitary surgery. *J Neurosurg.* 2020;134:1816-1823.
29. Pandit V, Lee A, Zeeshan M, et al. Effect of frailty syndrome on the outcomes of patients with carotid stenosis. *J Vasc Surg.* 2020;71:1595-1600.
30. Balasundaram N, Chandra I, Sunilkumar VT, Kanake S, Bath J, Vogel TR. Frailty index (mFI-5) predicts resource utilization after nonruptured endovascular aneurysm repair. *J Surg Res.* 2022;283:507-513.
31. Collins CE, Renshaw S, Adib M, Gupta A, Rosenthal R. Frailty in emergency general surgery: low-risk procedures pose similar risk as high-risk procedures for frail patients. *Surgery.* 2023;173:485-491.
32. Katz A, Alerte E, Akhavan A, et al. Can frailty indices predict surgical risk in open reduction and fixation of facial fractures? *J Craniofac Surg.* 2022;33:2573-2577.
33. Elsamadicy AA, Koo AB, Reeves BC, et al. Utilization of machine learning to model important features of 30-day readmissions following surgery for metastatic spinal column tumors: the influence of frailty. *Global Spine J.* 2022;1:2192568221138052.
34. Mehta T, Desai N, Patel S, et al. Readmission trends related to unruptured intracranial aneurysm treatment. *Front Neurol.* 2021;12:590751.
35. Kuwabara M, Ikawa F, Michihata N, et al. The 5-factor modified frailty index as a more useful associated factor than chronological age after unruptured cerebral aneurysm surgery: a nationwide registry study. *Neurosurgery.* 2023;92:329-337.
36. Kandregula S, Savardekar AR, Terrell D, et al. Microsurgical clipping and endovascular management of unruptured anterior circulation aneurysms: how age, frailty, and comorbidity indexes influence outcomes. *J Neurosurg.* 2022;30:1-11.
37. Caffes N, Wenger N, Cannarsa G, et al. Unruptured cerebral aneurysms in elderly patients: key challenges and management. *Ann Med.* 2021;53:1839-1849.

38. van der Vlies E, Smits AB, Los M, et al. Implementation of a preoperative multidisciplinary team approach for frail colorectal cancer patients: influence on patient selection, prehabilitation and outcome. *J Geriatr Oncol.* 2020;11:1237-1243.
39. Hampton JP, Owodunni OP, Bettick D, et al. Compliance to an enhanced recovery pathway among patients with a high frailty index after major gastrointestinal surgery results in improved 30-day outcomes. *Surgery.* 2019;166:75-81.
40. Theriault BC, Pazniokas J, Adkoli AS, et al. Frailty predicts worse outcomes after intracranial meningioma surgery irrespective of existing prognostic factors. *Neurosurg Focus.* 2020;49:E16.
41. Hoffman H, Jalal MS, Chin LS. A propensity score-matched comparison of readmission rates associated with microsurgical clipping and endovascular treatment of ruptured intracranial aneurysms. *J Stroke Cerebrovasc Dis.* 2020;29:104696.
42. Molyneux AJ. Changes in the treatment of patients with subarachnoid haemorrhage following publication of the International Subarachnoid Aneurysm Trial. *Clin Neurol Neurosurg.* 2006;108:115-116.
43. Molyneux AJ, Kerr RSC, Yu LM, et al. International subarachnoid aneurysm trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised comparison of effects on survival, dependency, seizures, rebleeding, subgroups, and aneurysm occlusion. *Lancet.* 2005;366:809-817.
44. Brinjikji W, Rabinstein AA, Lanzino G, Kallmes DF, Cloft HJ. Effect of age on outcomes of treatment of unruptured cerebral aneurysms: a study of the National Inpatient Sample 2001-2008. *Stroke.* 2011;42:1320-1324.
45. McDonald JS, McDonald RJ, Fan J, Kallmes DF, Lanzino G, Cloft HJ. Comparative effectiveness of unruptured cerebral aneurysm therapies: propensity score analysis of clipping versus coiling. *Stroke.* 2013;44:988-994.
46. Buta BJ, Walston JD, Godino JG, et al. Frailty assessment instruments: systematic characterization of the uses and contexts of highly-cited instruments. *Ageing Res Rev.* 2016;26:53-61.
47. Nidadavolu LS, Ehrlich AL, Sieber FE, Oh ES. Preoperative evaluation of the frail patient. *Anesth Analg.* 2020;130:1493-1503.

Conflict of interest statement: This work was supported by the National Center for Advancing Translational Science of the U.S. National Institutes of Health (grant UL1TR001855 to L.D.). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Received 13 October 2023; accepted 2 November 2023

Citation: *World Neurosurg.* (2024) 181:e882-e896.

<https://doi.org/10.1016/j.wneu.2023.11.005>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

1878-8750/\$ - see front matter © 2023 Elsevier Inc. All rights reserved.

SUPPLEMENTARY DATA

Supplementary Table 1. Inclusion Criteria for Study Cohort and Corresponding ICD-9 Codes

Disease Category: Unruptured Cerebral Aneurysm		ICD-9 Codes
Diagnosis code with description		
Cerebral aneurysm, nonruptured		437.3
Procedure code with description		
Clipping of aneurysm, other craniotomy/craniectomy		39.51, 01.24, 01.25
Endovascular embolization or occlusion, other endovascular procedures, other repair of aneurysm		39.72, 39.75, 39.76, 39.79, 39.52
Percutaneous insertion of intracranial vascular stent		00.65
ICD-9, International Classification of Diseases, 9th revision.		

Supplementary Table 2. ICD-Derived Frailty-Defining Diagnoses According to Johns Hopkins Adjusted Clinical Groups Method

Frailty-Defining Variable	Diagnosis	ICD-9 Codes
Malnutrition	Nutritional marasmus	261
	Other severe protein calorie malnutrition	262
Dementia	Senile dementia with delusional or depressive symptoms	290.20, 290.21
	Senile dementia with delirium	290.3
Severe visual impairment	Profound impairment, both eyes	369.00–369.08
	Moderate or severe impairment, better eye/lesser eye profound	369.10–369.18
Decubitus ulcer	Decubitus ulcer	707.00–707.09, 707.20–707.25
Urinary incontinence	Incontinence without sensory awareness	788.34
	Continuous leakage	788.37
Weight loss	Abnormal loss of weight and underweight	783.21, 783.22
	Feeding difficulties and mismanagement	783.3
Fecal incontinence	Incontinence of feces	787.60–787.63
Social support needs	Lack of housing	V60.0
	Inadequate housing	V60.1
	Inadequate material resources	V60.2
Walking difficulties	Difficulty in walking	719.7
	Abnormality of gait	781.2
Fall	Fall on stairs or steps	E880.0–E880.9
	Fall on wheelchair	E884.3
ICD-9, International Classification of Diseases, 9th revision.		

Supplementary Table 3. Surgical, Major Hospital, and Neurological Complication Diagnoses and ICD-9 Codes

Complication Diagnoses	ICD-9 Codes
Surgical	
Intracranial vascular complications	431, 99702
Intracranial hemorrhage	431
Iatrogenic cerebrovascular infarction or hemorrhage	99702
Mental status changes (delirium and transient mental disorders, alternation of consciousness, altered mental status)	2930, 2939, 78001, 78009, 78097
Hemorrhage or hematoma complicating a procedure	99811, 998.12
Postoperative nausea with or without vomiting	78701, 78702
Bacterial meningitis	3201–3209
Major	
Pneumonia	481–482, 482.1–482.3, 482.30–482.32, 482.39–482.41, 482.49, 482.80–482.84, 482.89, 482.90, 483.0, 483.1, 483.8, 485–487.0, 997.3, 507.0
Pulmonary embolism	415.1–415.9
Renal failure	584, 584.5–584.9
Cerebrovascular accident	433.01, 433.11, 433.21, 433.31, 433.81., 433.91
Myocardial infarction	410.00–410.90, 410.01, 410.11–410.91
Cardiac arrest	427.5
Sepsis	995.91
Septic shock	995.92
Neurological	
Intracerebral hemorrhage	431, 998.11–12
Subarachnoid hemorrhage	430
Transient cerebral ischemia	435
Seizures	345.xx
Neurological complications after procedure	997.01–997.09

ICD-9, International Classification of Diseases, 9th revision.